

Koryak Bayesian Pairwise Models: Inverse Inanimate Patients

Comparison in all four models: inverse 2-2 inanimate patient vs inverse 2-1 inanimate patient. Coding is +1 for inverse 2-2 inanimate and -1 for inverse 2-1 inanimate; contrast terms are therefore half-differences, and the full-difference column reports 2*beta.

Models: empirical-logit fixation proportions with cubic time, condition contrast, and contrast-by-time interactions; fit separately for agent and patient looks in LE1 and LE3.

Data Included

Window	Condition	Sentences included	Participants	50 ms bins
LE1	Inverse 2-1, inanimate patient	93	16	1964
LE1	Inverse 2-2, inanimate patient	74	14	1357
LE3	Inverse 2-1, inanimate patient	93	16	1953
LE3	Inverse 2-2, inanimate patient	74	14	1554

Totals: LE1 included 167 sentences; LE3 included 167 sentences.

Credible Contrast Terms

Rows shown here are contrast or contrast-by-time terms whose 95% credible interval excludes zero.

Window	Looks	Term	Direction	Beta	95% CI	P(beta > 0)
LE1	agent looks	Time 1 x contrast	positive	22.122	[7.231, 37.128]	0.998
LE1	agent looks	Time 3 x contrast	positive	5.408	[0.316, 10.554]	0.981
LE1	patient looks	Time 1 x contrast	negative	-17.872	[-33.216, -2.920]	0.011
LE1	patient looks	Time 2 x contrast	positive	11.512	[5.255, 17.793]	1.000
LE1	patient looks	Time 3 x contrast	negative	-6.538	[-12.938, -0.322]	0.020
LE3	agent looks	Time 2 x contrast	positive	7.793	[2.194, 13.318]	0.998

Contrast And Interaction Terms

Window	Looks	Term	Beta	95% CI	P(beta > 0)	Full difference 2*beta
LE1	agent looks	Main contrast	-0.140	[-0.832, 0.540]	0.336	-0.280 [-1.665, 1.080]
LE1	agent looks	Time 1 x contrast	22.122	[7.231, 37.128]	0.998	44.244 [14.463, 74.255]
LE1	agent looks	Time 2 x contrast	-2.087	[-7.485, 3.276]	0.223	-4.175 [-14.971, 6.553]
LE1	agent looks	Time 3 x contrast	5.408	[0.316, 10.554]	0.981	10.815 [0.632, 21.109]
LE1	patient looks	Main contrast	0.168	[-0.730, 1.023]	0.652	0.335 [-1.459, 2.047]
LE1	patient looks	Time 1 x contrast	-17.872	[-33.216, -2.920]	0.011	-35.744 [-66.432, -5.840]
LE1	patient looks	Time 2 x contrast	11.512	[5.255, 17.793]	1.000	23.023 [10.510, 35.587]

Window	Looks	Term	Beta	95% CI	P(beta > 0)	Full difference 2*beta
LE1	patient looks	Time 3 x contrast	-6.538	[-12.938, -0.322]	0.020	-13.076 [-25.876, -0.645]
LE3	agent looks	Main contrast	-0.310	[-0.892, 0.290]	0.148	-0.619 [-1.784, 0.580]
LE3	agent looks	Time 1 x contrast	2.748	[-9.307, 15.723]	0.661	5.496 [-18.613, 31.445]
LE3	agent looks	Time 2 x contrast	7.793	[2.194, 13.318]	0.998	15.586 [4.389, 26.635]
LE3	agent looks	Time 3 x contrast	-1.315	[-6.729, 4.185]	0.315	-2.630 [-13.457, 8.369]
LE3	patient looks	Main contrast	0.446	[-0.322, 1.215]	0.877	0.892 [-0.645, 2.430]
LE3	patient looks	Time 1 x contrast	0.805	[-15.859, 17.788]	0.531	1.609 [-31.718, 35.576]
LE3	patient looks	Time 2 x contrast	-0.610	[-6.193, 5.065]	0.413	-1.220 [-12.387, 10.130]
LE3	patient looks	Time 3 x contrast	1.960	[-3.916, 7.729]	0.753	3.919 [-7.831, 15.458]

All Fixed Effects

Window	Looks	Term	Beta	95% CI	P(beta > 0)	Full difference 2*beta
LE1	agent looks	Intercept	1.134	[0.385, 1.902]	0.997	
LE1	agent looks	Time 1	35.309	[19.138, 52.193]	1.000	
LE1	agent looks	Time 2	1.563	[-3.880, 6.895]	0.718	
LE1	agent looks	Time 3	-8.507	[-13.644, -3.259]	0.001	
LE1	patient looks	Intercept	-2.368	[-3.410, -1.358]	0.000	
LE1	patient looks	Time 1	-42.366	[-59.278, -25.702]	0.000	
LE1	patient looks	Time 2	-2.652	[-8.796, 3.549]	0.197	
LE1	patient looks	Time 3	8.763	[2.533, 15.022]	0.996	
LE3	agent looks	Intercept	0.508	[-0.239, 1.286]	0.907	
LE3	agent looks	Time 1	-20.474	[-34.992, -5.616]	0.006	
LE3	agent looks	Time 2	-7.107	[-12.326, -1.766]	0.007	
LE3	agent looks	Time 3	-1.764	[-7.473, 3.790]	0.261	
LE3	patient looks	Intercept	-1.397	[-2.424, -0.408]	0.005	
LE3	patient looks	Time 1	19.501	[-3.095, 41.729]	0.954	
LE3	patient looks	Time 2	-2.683	[-8.210, 2.986]	0.164	
LE3	patient looks	Time 3	10.590	[5.063, 16.176]	1.000	
LE1	agent looks	Main contrast	-0.140	[-0.832, 0.540]	0.336	-0.280 [-1.665, 1.080]
LE1	agent looks	Time 1 x contrast	22.122	[7.231, 37.128]	0.998	44.244 [14.463, 74.255]
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Convergence Diagnostics

All four models had 0 divergent transitions and 0 max-treedepth hits. Fixed-effect Rhats are approximately 1.00.

Window	Looks	Max Rhat	Min bulk ESS	Min tail ESS	Worst bulk variable	Worst tail variable	Divergences	Treedepth hits
LE1	agent looks	1.0106	612	1661	sd_participant__Intercept	lp__	0	0
LE1	patient looks	1.0076	611	910	sd_participant__Intercept	sd_participant__Intercept	0	0
LE3	agent looks	1.0049	774	1194	sd_participant__Intercept	sd_participant__Intercept	0	0
LE3	patient looks	1.0071	647	1273	sd_participant__time	sd_participant__time	0	0

Fixed-Effect Diagnostics

Window	Looks	Term	Rhat	Bulk ESS	Tail ESS
LE1	agent looks	Intercept	1.0017	1600	2344
LE1	agent looks	Time 1	1.0016	1554	2041
LE1	agent looks	Time 2	1.0024	8152	3030
LE1	agent looks	Time 3	0.9999	7731	3320
LE1	agent looks	Main contrast	1.0017	1297	1864
LE1	agent looks	Time 1 x contrast	1.0011	1401	1935
LE1	agent looks	Time 2 x contrast	1.0011	7113	2674
LE1	agent looks	Time 3 x contrast	1.0010	7603	3111
LE1	patient looks	Intercept	1.0032	1657	2185
LE1	patient looks	Time 1	1.0009	2043	2615
LE1	patient looks	Time 2	1.0011	7875	3089
LE1	patient looks	Time 3	1.0011	7474	2306
LE1	patient looks	Main contrast	1.0019	1510	2242
LE1	patient looks	Time 1 x contrast	1.0007	1796	2493
LE1	patient looks	Time 2 x contrast	1.0011	6564	2871

Window	Looks	Term	Rhat	Bulk ESS	Tail ESS
LE1	patient looks	Time 3 x contrast	1.0006	7006	2923
LE3	agent looks	Intercept	1.0013	1774	2351
LE3	agent looks	Time 1	1.0008	2335	2520
LE3	agent looks	Time 2	1.0018	8210	2981
LE3	agent looks	Time 3	0.9997	7622	2946
LE3	agent looks	Main contrast	1.0030	1311	2145
LE3	agent looks	Time 1 x contrast	1.0025	2073	2673
LE3	agent looks	Time 2 x contrast	1.0021	10647	3131
LE3	agent looks	Time 3 x contrast	1.0024	8133	2769
LE3	patient looks	Intercept	1.0015	1890	2272
LE3	patient looks	Time 1	1.0007	2472	2836
LE3	patient looks	Time 2	1.0007	7334	3257
LE3	patient looks	Time 3	1.0008	7422	3145
LE3	patient looks	Main contrast	1.0012	1785	2146
LE3	patient looks	Time 1 x contrast	1.0014	2162	2064
LE3	patient looks	Time 2 x contrast	1.0010	7912	3250
LE3	patient looks	Time 3 x contrast	0.9999	7849	3127